

Analytic Evaluation of QED Corrections in One-Electron Ions

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Outline

- 1 Motivation
- 2 One-Loop QED Corrections
- 3 Vacuum Polarization Correction
 - Leptonic
 - Hadronic
- 4 Self-Energy Correction to Energy Levels
- 5 Conclusion and Outlook

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Motivation

- Theory of Everything
- Testing Current Fundamental Theories
- Bound-State Quantum Electrodynamics
- Highly Charged Ions
- High-Precision Experiments

⇒ High-Precision Theory!

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Quantum Electrodynamics (QED)

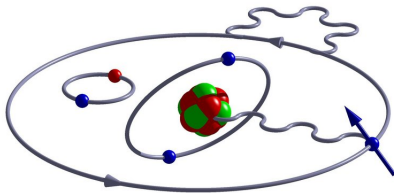


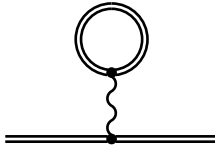
Figure: Scheme of the QED contributions to the electronic structure of highly charged ions.¹

$$\mathcal{L}_{\text{QED}} = \bar{\psi} \left[\gamma^\mu (i\hbar c \partial_\mu - e A_\mu) - m_e c^2 \right] \psi - \frac{1}{16\pi} F_{\mu\nu} F^{\mu\nu}. \quad (1)$$

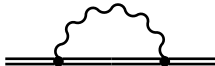
¹ https://www.mpi-hd.mpg.de/mpi/fileadmin/bilder/Progress_Reports/2017-19/2QuantumDynamics.pdf

One-Loop Energy Corrections

Vacuum Polarization:

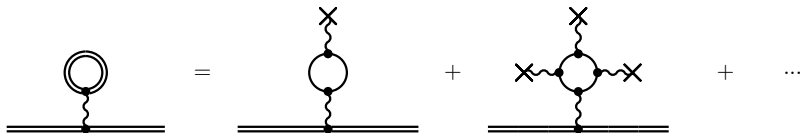


Self-Energy Correction:



Vacuum Polarization

Vacuum Polarization:



Modification of the Photon Propagator:

$$\begin{aligned}
 iD'_{\mu\nu}(k) &= \text{wavy line} + \text{wavy line} \text{---} \text{circle} \text{---} \text{wavy line} \\
 &= iD_{\mu\nu}(k) + iD_{\mu\lambda}(k) \frac{i\Pi^{\lambda\sigma}(k)}{4\pi} iD_{\sigma\nu}(k)
 \end{aligned}$$

One-Loop g Factor Corrections

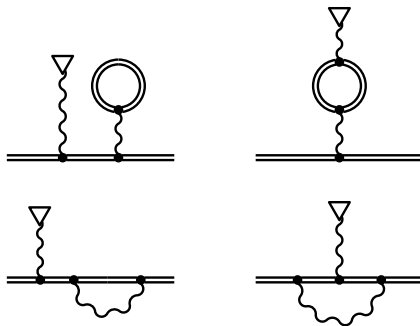


Figure: Feynman diagrams representing the first-order radiative corrections to the g factor of the bound electron.

g Factor

The g Factor is given by [18]:

$$g = -\frac{\kappa}{2j(j+1)} \left(1 - 2\kappa \frac{\partial E_{n\kappa}}{\partial m_e} \right), \quad (2)$$

if the potential $V(r)$ does not depend on the electron mass m_e .

Small Perturbation of the potential $\delta V(r)$ leads to

$$\Delta g = -\frac{\kappa^2}{j(j+1)m_e} \left\langle r \frac{\partial \delta V(r)}{\partial r} \right\rangle. \quad (3)$$

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Leptonic Vacuum Polarization

Leptonic Uehling Potential [9]:

$$\delta V(r) = \frac{\alpha}{\pi} \int_0^1 dv \frac{v^2 (1 - v^2/3)}{1 - v^2} \left(-\frac{Z\alpha}{r} e^{-2m_l r / \sqrt{1-v^2}} \right), \quad (4)$$

where m_l is the mass of the virtual particle in the fermionic loop.

Energy Shift of the 1s State:

$$\Delta E_{1s}^{\text{lept. VP}} = \langle \delta V(r) \rangle_{1s} = \int_0^\infty dr (G_{1s}^2(r) + F_{1s}^2(r)) \delta V(r). \quad (5)$$

g Factor Shift of the 1s State:

$$\Delta g_{1s}^{\text{lept. VP}} = -\frac{4}{3m_e} \left\langle r \frac{\partial \delta V(r)}{\partial r} \right\rangle_{1s}. \quad (6)$$

4.1. Energy Shift

We now perform the calculations for the energy shift of the $1s$ state in hydrogen-like atoms due to the leptonic Uehling potential. When $|\psi_{1s}\rangle$ denotes the bound electron wave function of the ground state in a point-like Coulomb potential, the energy shift in first-order perturbation theory is given by:

$$\begin{aligned}\Delta E_{1s}^{\text{VP}} &= \langle \psi_{1s} | \delta V(r) | \psi_{1s} \rangle = \int_0^\infty dr (G^2(r) + F^2(r)) \delta V(r) \\ &= \frac{2\lambda}{\Gamma(2\gamma+1)} \int_0^\infty dr (2\lambda r)^{2\gamma} e^{-2\lambda r} \left(-\frac{Z\alpha^2}{4\pi} \int_0^\infty dv \frac{v^2 (1-v^2/3)}{1-v^2} e^{-2m_e r/\sqrt{1-v^2}} \right) \\ &= -\frac{2\lambda Z\alpha^2}{\Gamma(2\gamma+1)\pi} \int_0^\infty dv \frac{v^2 (1-v^2/3)}{1-v^2} \int_0^\infty dr \frac{(2\lambda r)^{2\gamma}}{r} e^{-2r \left(\lambda + \frac{m_e}{\sqrt{1-v^2}} \right)}.\end{aligned}\quad (4.4)$$

Now, substitute $u = r^{2\gamma}$, $du = 2\gamma r^{2\gamma-1} dr$.

and then $t = \left[2 \left(\lambda + \frac{m_e}{\sqrt{1-v^2}} \right) \right]^{2\gamma} u$, $dt = \left[2 \left(\lambda + \frac{m_e}{\sqrt{1-v^2}} \right) \right]^{2\gamma} du$:

$$\begin{aligned}\Delta E_{1s}^{\text{VP}} &= -\frac{2\lambda Z\alpha^2}{\Gamma(2\gamma+1)\pi} \int_0^1 dv \frac{v^2 (1-v^2/3)}{1-v^2} \int_0^\infty \frac{du}{2\gamma} (2\lambda)^{2\gamma} e^{-2\lambda u/\sqrt{1-v^2}} \left(\lambda + \frac{m_e}{\sqrt{1-v^2}} \right) \\ &= -\frac{2\lambda Z\alpha^2}{\Gamma(2\gamma+1)\pi} \int_0^1 dv \frac{v^2 (1-v^2/3)}{1-v^2} \frac{(2\lambda)^{2\gamma}}{2\gamma \left[2 \left(\lambda + \frac{m_e}{\sqrt{1-v^2}} \right) \right]^{2\gamma}} \int_0^\infty dt e^{-t} t^{\frac{1}{2\gamma}-1} \\ &= -\frac{Z\alpha^2 \lambda^{2\gamma+1}}{\gamma\pi} \int_0^1 dv \frac{v^2 (1-v^2/3) (1-v^2)^{\gamma-1}}{\left[\lambda\sqrt{1-v^2} + m_e \right]^{2\gamma}} \\ &= -\frac{Z\alpha^2 (sZ\alpha)^2 \lambda}{\gamma\pi} \int_0^1 dv \frac{v^2 (1-v^2/3) (1-v^2)^{\gamma-1}}{\left[1 + sZ\alpha\sqrt{1-v^2} \right]^{2\gamma}},\end{aligned}\quad (4.5)$$

where we used the relation $\Gamma(x+1) = x\Gamma(x)$ and that $\lambda/m_e = sZ\alpha$, where $s = m_e/m_l$ is the ratio of the electron and the loop particle masses.

This integral can now be solved analytically with the base integral given in [28]:

$$\begin{aligned}I_{abc} &= \int_0^1 dy \frac{(1-y^2)^{a-1/2}}{y^{b-1}} \left(\frac{sZ\alpha y}{1+sZ\alpha y} \right)^{c-2a} \\ &= \frac{1}{2} (sZ\alpha)^{c-2a} B \left(a + \frac{1}{2}, 1 - \frac{b-c}{2} - \epsilon \right) \\ &\quad \times {}_3F_2 \left(\frac{c}{2} - \epsilon, \frac{c+1}{2} - \epsilon, 1 - \frac{b-c}{2} - \epsilon; \frac{1}{2}, a + \frac{3-b+c}{2} - \epsilon; (sZ\alpha)^2 \right) \\ &\quad - \frac{c-2\epsilon}{2} (sZ\alpha)^{c+1-2a} B \left(a + \frac{1}{2}, \frac{3-b+c}{2} - \epsilon \right) \\ &\quad \times {}_3F_2 \left(\frac{c}{2} + 1 - \epsilon, \frac{c+1}{2} - \epsilon, \frac{3-b+c}{2} - \epsilon; \frac{3}{2}, a + 2 - \frac{b-c}{2} - \epsilon; (sZ\alpha)^2 \right).\end{aligned}\quad (4.6)$$

Here, $\epsilon = 1 - \gamma$, and $B(x, y)$ is the beta function defined in Appendix A.4. Performing the substitution $y = \sqrt{1-v^2}$, $dy = \frac{v dv}{\sqrt{1-v^2}}$, one obtains finally:

$$\begin{aligned}\Delta E_{1s}^{\text{VP}} &= -\frac{Z\alpha^2 (sZ\alpha)^{2\gamma} \lambda}{\gamma\pi} \left[\int_0^1 \frac{y dy}{\sqrt{1-y^2}} \frac{(1-y^2)^{\gamma-3/2}}{[1+sZ\alpha y]^{2\gamma}} - \frac{1}{3} \int_0^1 \frac{y dy}{\sqrt{1-y^2}} \frac{(1-y^2)^2 y^{\gamma-2}}{[1+sZ\alpha y]^{2\gamma}} \right] \\ &= -\frac{Z\alpha^2 \lambda}{\gamma\pi} \left[\int_0^1 dy \frac{(1-y^2)^{\frac{\gamma}{2}}}{y} \left(\frac{sZ\alpha y}{1+sZ\alpha y} \right)^{2\gamma} - \int_0^1 dy \frac{(1-y^2)^{\frac{\gamma}{2}}}{y} \left(\frac{sZ\alpha y}{1+sZ\alpha y} \right)^{2\gamma} \right] \\ &= -\frac{Z\alpha^2 \lambda}{\gamma\pi} \left[I_{122} - \frac{1}{3} I_{222} \right].\end{aligned}\quad (4.7)$$

This result was already derived in [27] and reference therein. The difference is that in [27], the result was derived for the electron as loop-particle and electrons or muons as bound particles. Here, we considered the electron as bound particle and arbitrary leptons as loop-particles.

To investigate this all-order in $Z\alpha$ result, we may expand (4.7) in a series for small $Z\alpha$ to get a simple and handy formula for the correction due to the leptonic Uehling potential. Using the computer algebra software *Mathematica* [18], we find for the leading order Taylor expansion:

$$\begin{aligned}\Delta E_{1s}^{\text{VP}} &= \frac{\alpha m_e}{\pi} \left[-\frac{4s^2(Z\alpha)^4}{15} + \frac{5\pi s^3(Z\alpha)^5}{48} \right. \\ &\quad + \left(\frac{4s^2}{15} \ln(2sZ\alpha) - \frac{107s^2}{225} - \frac{12s^4}{35} \right) (Z\alpha)^6 \\ &\quad \left. + \left(\frac{5\pi s^3}{48} \ln \left(\frac{sZ\alpha}{2} \right) - \frac{17\pi s^3}{576} + \frac{7\pi s^5}{64} \right) (Z\alpha)^7 + \mathcal{O}((Z\alpha)^8) \right].\end{aligned}\quad (4.8)$$

Therefore, the energy shift correction for leptonic vacuum polarization is of the leading order $(Z\alpha)^4$. The first term of expression (4.8) coincides with the result of the simplest approximation, namely, using nonrelativistic wave functions and the low-momentum approximation of the polarization function [13, 16, 24]. Since this leads to a Dirac delta function, we will call it the δ -potential approximation.

With formula (4.7), one can calculate the energy shift contribution for any leptonic vacuum loop.² As equation (4.7) or (4.8) show, the heavier the loop particle is, the smaller is the measurable effect. After the e^+e^- -loop, the next important contribution is due to the $\mu^+\mu^-$ -loop, also called muonic vacuum polarization, where a muon and anti-muon pair is produced in the loop. The effect of muonic vacuum polarization is around $s^2 \approx 1/207^2 \approx 2 \cdot 10^{-5}$ times smaller than that of the electronic vacuum polarization.

Figure 4.1 shows the absolute value of the muonic vacuum polarization contribution to the energy shift for different charge numbers Z , calculated with the exact formula (4.7) and its approximation (4.8) up to 5th order. It should be mentioned that the Uehling potential is attractive and results in a stronger binding. Thus, the corresponding energy shift is negative. The correction for ions with charge numbers $Z \leq 8$ is so small that its value is not visible in this diagram.

²As mentioned in [28], this formula can be applied to regular, electronic atoms or ions considered in this work, or to muonic atoms by replacing the electron mass m_e with the muon mass m_μ .

Leptonic Vacuum Polarization

Energy Shift for a point-like Nucleus:

$$\Delta E_{1s}^{\text{lept. VP}} = -\frac{Z\alpha^2\lambda}{\gamma\pi} \left[l_{122} - \frac{1}{3} l_{222} \right]. \quad (7)$$

g Factor Shift for a point-like Nucleus:

$$\Delta g_{1s}^{\text{lept. VP}} = -\frac{8\alpha(Z\alpha)}{3\pi s} \left[l_{133} - \frac{1}{3} l_{233} + \frac{Z\alpha s}{2\gamma} \left(l_{122} - \frac{1}{3} l_{222} \right) \right]. \quad (8)$$

Hadronic Vacuum Polarization

Real Part of the Hadronic Polarization Function [4]:

$$\text{Re} [\Pi_{\text{had}}(q^2)] = A_i + B_i \ln(1 + C_i q^2), \quad (9)$$

The parameters are given by

i	Region	Range [GeV]	A_i	B_i	$C_i [\text{GeV}^{-2}]$
1	$0 - k_1$	$0.0 - 0.7$	0.0	0.0023092	3.9925370
2	$k_1 - k_2$	$0.7 - 2.0$	0.0	0.0022333	4.2191779
3	$k_2 - k_3$	$2.0 - 4.0$	0.0	0.0024402	3.2496684
4	$k_3 - k_4$	$4.0 - 10.0$	0.0	0.0027340	2.0995092
5	$k_4 - k_5$	$10.0 - m_Z$	0.0010485	0.0029431	1.0
6	$k_5 - k_6$	$m_Z - 10^4$	0.0012234	0.0029237	1.0
7	$k_5 - k_6$	$10^4 - 10^5$	0.0016894	0.0028984	1.0

Hadronic Uehling Potential

Numerical Hadronic Uehling Potential for a point-like Nucleus:

$$\delta V_{\text{numerical}}^{\text{had. VP}}(r) = -\frac{2Z\alpha}{\pi} \sum_{k=1}^7 \int_{k_{i-1}}^{k_i} dq \frac{\sin(qr)}{qr} [A_i + B_i \ln(1 + C_i q^2)] . \quad (10)$$

Analytical Hadronic Uehling Potential for a point-like Nucleus:

$$\begin{aligned} \delta V_{\text{analytical}}^{\text{had. VP}}(r) &= -\frac{2Z\alpha}{\pi} \int_0^\infty dq \frac{\sin(qr)}{qr} [A_1 + B_1 \ln(1 + C_1 q^2)] \\ &= -\frac{2Z\alpha}{r} B_1 E_1 \left(\frac{r}{\sqrt{C_1}} \right) . \end{aligned} \quad (11)$$

Hadronic Uehling Potential

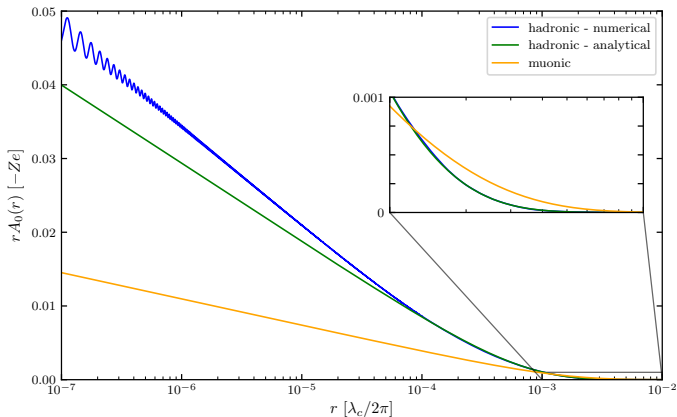


Figure: Comparison of leptonic and hadronic Uehling potential.

Hadronic Energy Shift

Energy Shift of the 1s State for a point-like Nucleus:

$$\Delta E_{1s}^{\text{had. VP}} = -\frac{Z\alpha\lambda(2\lambda\sqrt{C_1})^{2\gamma}B_1}{\gamma^2} {}_2F_1\left(2\gamma, 2\gamma; 1+2\gamma; -2\lambda\sqrt{C_1}\right). \quad (12)$$

$Z\alpha$ Expansion:

$$\begin{aligned} \Delta E_{1s}^{\text{had. VP}} \approx & -4B_1C_1m_e^3(Z\alpha)^4 + \frac{32B_1C_1^{3/2}m_e^4(Z\alpha)^5}{3} \\ & -4B_1C_1m_e^3(Z\alpha)^6 \left[1 + 6C_1m_e^2 - \ln(2Z\alpha\sqrt{C_1}m_e)\right]. \end{aligned} \quad (13)$$

Hadronic Energy Shift

Z	$\Delta E_{\text{analytical}}^{\text{point}} [\text{eV}]$	$\Delta E_{\text{numerical}}^{\text{point}} [\text{eV}]$	$\Delta E_{\text{exact}}^{\text{finite size}} [\text{eV}]$
1	$-1.3963 \cdot 10^{-11}$	$-1.39(33) \cdot 10^{-11}$	$-1.391(4) \cdot 10^{-11}$
14	$-5.9178 \cdot 10^{-7}$	$-5.90(18) \cdot 10^{-7}$	$-5.756(1) \cdot 10^{-7}$
20	$-2.7133 \cdot 10^{-6}$	$-2.71(5) \cdot 10^{-6}$	$-2.5596(3) \cdot 10^{-6}$
70	$-3.1090 \cdot 10^{-3}$	$-3.109(4) \cdot 10^{-3}$	$-1.248(1) \cdot 10^{-3}$
82	$-1.4128 \cdot 10^{-2}$	$-1.413(1) \cdot 10^{-2}$	$-3.693(4) \cdot 10^{-3}$

Table: Energy shifts for hadronic VP.

Hadronic g Factor Shift

g Factor Shift of the $1s$ State for a point-like Nucleus:

$$\Delta g_{1s}^{\text{had. VP}} = -\frac{8B_1(Z\alpha)^2(2\lambda\sqrt{C_1})^{2\gamma}}{3\gamma(1+2\lambda\sqrt{C_1})^{2\gamma}} + \frac{4}{3m_e}\Delta E_{\text{approx.}} \quad (14)$$

$Z\alpha$ Expansion:

$$\begin{aligned} \Delta g_{1s}^{\text{had. VP}} \approx & -16B_1C_1m_e^2(Z\alpha)^4 + \frac{512B_1C_1^{3/2}m_e^3(Z\alpha)^5}{9} \\ & - \frac{16B_1C_1m_e^2(Z\alpha)^6}{3} \left[2 + 30C_1m_e^2 - 3\ln(2m_eZ\alpha\sqrt{C_1}) \right]. \end{aligned} \quad (15)$$

Hadronic g Factor Shift

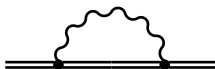
Z	$\Delta g_{\text{analytical}}^{\text{point}}$	$\Delta g_{\text{numerical}}^{\text{point}}$	$\Delta g_{\text{approx}}^{\text{finite size}}$
1	$-1.0929 \cdot 10^{-16}$	$-1.09(9) \cdot 10^{-16}$	$-1.09(2) \cdot 10^{-16}$
14	$-4.6157 \cdot 10^{-12}$	$-4.61(5) \cdot 10^{-12}$	$-4.49(1) \cdot 10^{-12}$
20	$-2.1085 \cdot 10^{-11}$	$-2.11(2) \cdot 10^{-11}$	$-1.99(1) \cdot 10^{-11}$
70	$-2.2051 \cdot 10^{-8}$	$-2.205(1) \cdot 10^{-8}$	$-8.86(1) \cdot 10^{-9}$
82	$-9.5886 \cdot 10^{-8}$	$-9.589(3) \cdot 10^{-8}$	$-2.51(1) \cdot 10^{-8}$

Table: g factor shifts for hadronic VP.

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Self-Energy Correction



$$\Delta E^{\text{SE}} = -ie^2 \int \frac{d^4 k}{(2\pi)^4} \frac{1}{k^2} \left\langle \psi \left| \gamma^\mu \frac{1}{\not{p} - \not{k} - m_e - \gamma^0 V} \gamma_\mu \right| \psi \right\rangle. \quad (16)$$

- Divergent Expression
- Exact Coulomb-Dirac Propagator only known in Coordinate Space
- Evaluation of the above Expression difficult

⇒ Approximation!

Dimensional Regularization

Idea: Integrals in QFT are only divergent in 3 or 4 dimensions!

⇒ Analytical Continuation to $d = 3 - 2\varepsilon$ and $D = 4 - 2\varepsilon$.

⇒ Taylor Expansion for small ε .

This Regularization Technique yields the well-known Result [9]

$$\Delta E_{1s}^{\text{SE}} = \frac{4\alpha}{3\pi} (Z\alpha)^4 m_e \left[\frac{5}{6} - 2 \ln(Z\alpha) - \ln k_0 \right], \quad (17)$$

where the Bethe logarithm $\ln k_0$ is defined as

$$\frac{(Z\alpha)^4 m_e}{n^3} \ln k_0 = \frac{1}{2m_e^2} \left\langle \mathbf{p} (H - E) \ln \left[\frac{2(H - E)}{m_e (Z\alpha)^2} \right] \mathbf{p} \right\rangle. \quad (18)$$

Bethe Logarithm

Integral representation of the Bethe logarithm [13]:

$$\ln k_0(n) = -\frac{3}{4} PV \int_0^1 dt \frac{1}{t^3} \left(\frac{t^2 - 1}{nt^2} P_{nd}(t) + \frac{2}{3n} - \frac{8t^2}{3} \right) - 2 \ln(n), \quad (19)$$

PV denotes the principal value and $P_{nd}(t)$ is the non-relativistic dipol matrix element:

$$P_{nd} = \frac{1}{3m_e} \left\langle \phi_{nlm} \left| \mathbf{p} \frac{1}{H - (E - \omega)} \mathbf{p} \right| \phi_{nlm} \right\rangle. \quad (20)$$

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Conclusion

- Fundamental Properties of the Bound Electron
- Analytic Evaluation of One-Loop QED Corrections
- Leptonic Vacuum Polarization
- Hadronic Vacuum Polarization
- Dimensional Regularization of One-Loop Self-Energy
- Analytic Evaluation of $1s$ Bethe logarithm

Outlook

- Hadronic Energy Shift (Paper with J. S. Breidenbach)
- Hadronic g Factor Shift for Extended Nucleus (Paper)
- Nucleus Models
- Bethe logarithm for $2s$ state
- Bethe logarithm for general ns state

⇒ Plenty of Room for Analytical Work!

Thank you and stay healthy!



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